

Question 1

1. Divide-and-Conquer : break problem into small, manageable tasks
2. Encapsulation & Modularity : Each object knows how to solve its task and has the information it needs.
3. Public interface : each object should present a clear public interface that determines how other objects will be used.
4. Information hiding : each object should shield its users from unnecessary details of how it performs its role.
5. Generality : Object should be designed to be as general as possible
6. Extensibility : An object should be designed so that its functionality can be extended to carry out more specialized tasks.
7. Abstraction : Focus only on those attributes that are essential for solving a particular problem.

Question 2

1. Save development time and cost by reusing code
2. easier debugging